



Year 12
Student Number

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2020 HSC Trial Examination

Year 12 Mathematics Extension 1

Examiner: AC

Reading time – 10 minutes

Writing time – 2 hours

General Instructions

- Write using black or blue pen.
- Read the instructions carefully – you are required to answer the questions in the space provided.
- If you use booklets, start each question in a separate writing booklet.
- Write your student name clearly on each page.
- Board-approved calculators may be used, unless stated otherwise.
- All diagrams must be drawn in pencil.
- Do not remove this question paper from the examination room.

Section	Guidance	Marks Available	Your Score
SECTION I	• Type of Questions – Multiple Choice • Attempt Questions 1 - 10 • Timing 15 minutes	10	
SECTION II	• Type of Questions – Multiple Choice • Attempt Questions 11 - 14 • Timing 1 hour 45 minutes	60	
Totals		70	

FINAL MARK	/ 70	%
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Your examination paper begins overleaf.

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Section 1: 10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

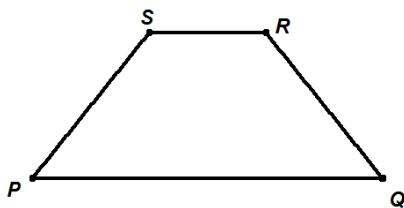
- 1 Given $(x - 2)$ is a factor of $x^3 - 8x^2 + 21x - A$, which of the following is the value of A ?

- (A) $A = -82$
- (B) $A = -2$
- (C) $A = 2$
- (D) $A = 18$

- 2 Which of the following is the derivative of $\tan^{-1}(3x)$?

- (A) $3 \tan^{-1} x$
- (B) $\frac{3}{1+x^2}$
- (C) $\frac{3}{1+9x^2}$
- (D) $3 \sec^2 3x$

- 3 $PQRS$ is a trapezium where $\overrightarrow{PS} = \vec{p}$, $\overrightarrow{SR} = \vec{s}$ and $\overrightarrow{PQ} = 2 \overrightarrow{SR}$.

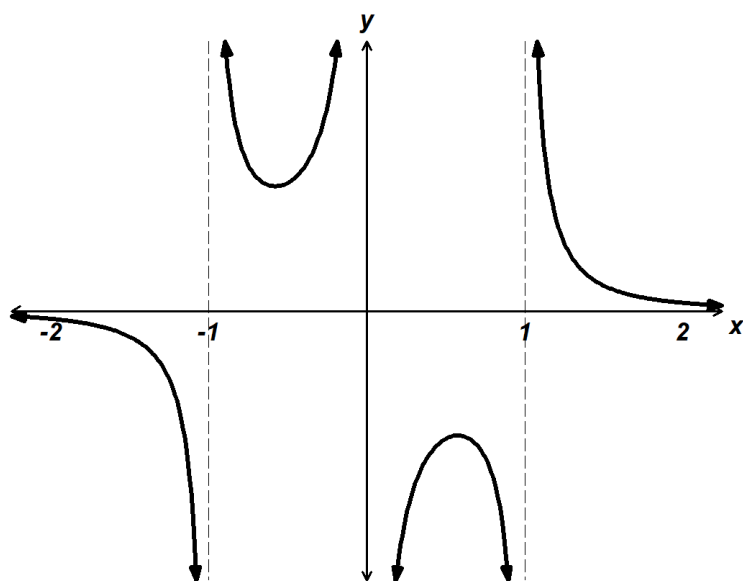


Which of the following is equivalent to $\overrightarrow{QS}$?

- (A) $2 \vec{s} + \vec{p}$
- (B) $2 \vec{s} - \vec{p}$
- (C) $\vec{p} - 2 \vec{s}$
- (D) $-\vec{p} - 2 \vec{s}$

- 4 Which of the following is the coefficient of x^4 in the expansion $\left(x + \frac{3}{x}\right)^8$?
- (A) 28
 (B) 56
 (C) 84
 (D) 252

5

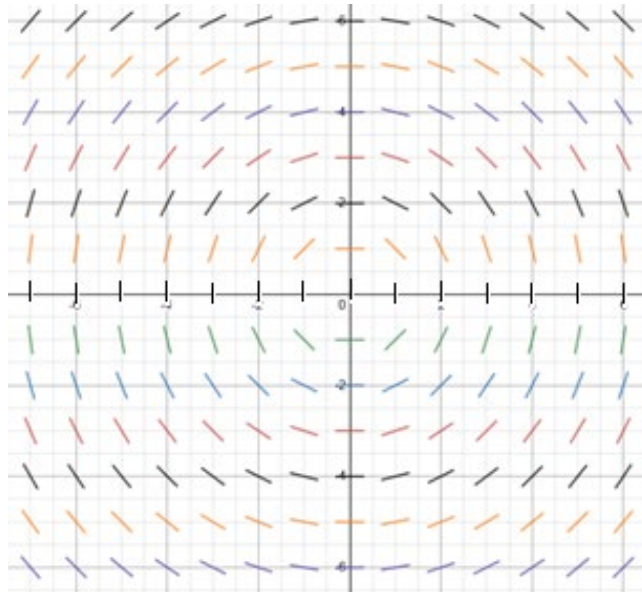


The graph above shows $y = \frac{1}{f(x)}$.

Which of the equations below best represents $f(x)$?

- (A) $f(x) = x^2 - 1$
 (B) $f(x) = x(x^2 - 1)$
 (C) $f(x) = x^2(x^2 - 1)$
 (D) $f(x) = x^2(x^2 - 1)^2$

- 6 The slope field for a first order differential equation is shown below.



Which of the following could be the differential equation represented?

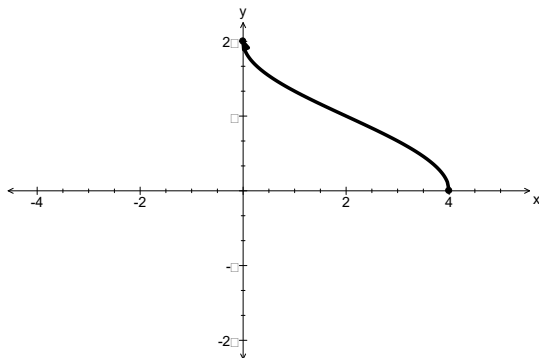
- (A) $\frac{dy}{dx} = \frac{x}{y}$
- (B) $\frac{dy}{dx} = \frac{-x}{y}$
- (C) $\frac{dy}{dx} = xy$
- (D) $\frac{dy}{dx} = -xy$

- 7 Four female and four male students are to be seated around a circular table.
In how many ways can this be done if the males and females must alternate?

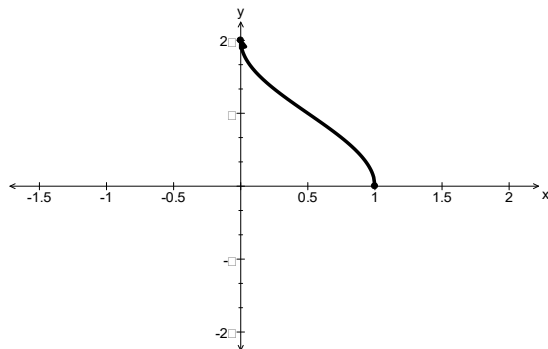
- (A) $4! \times 4!$
- (B) $3! \times 4!$
- (C) $3! \times 3!$
- (D) $2 \times 3! \times 3!$

8 Which of the graphs below shows $y = 2 \cos^{-1}\left(\frac{x}{2} - 1\right)$?

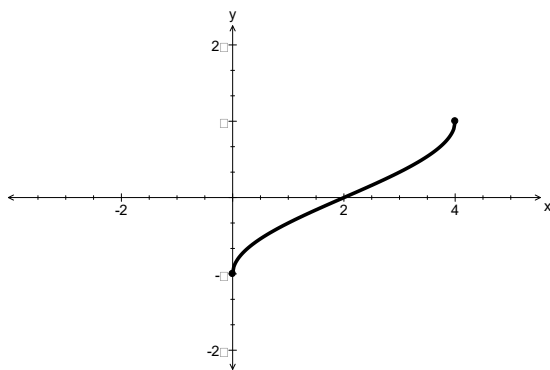
(A)



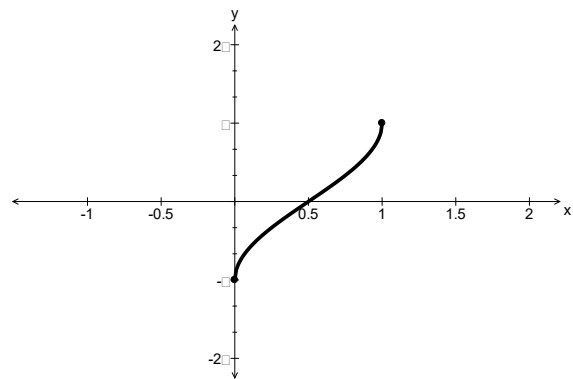
(B)



(C)



(D)



9 Which of the following expressions represents the area of the region bounded by the curve $y = \sin^3 x$ and the x -axis from $x = -\pi$ to $x = 2\pi$? Use the substitution $u = \cos x$.

(A) $-\int_{-\pi}^{2\pi} (1 - u^2) du$

(B) $-3 \int_0^{\pi} (1 - u^2) du$

(C) $-\int_{-1}^1 (1 - u^2) du$

(D) $3 \int_{-1}^1 (1 - u^2) du$

- 10 Emma made an error proving that $2^n + (-1)^{n+1}$ is divisible by 3 for all integers $n \geq 1$ using mathematical induction. The proof is shown below.

Step 1: To prove $2^n + (-1)^{n+1}$ is divisible by 3 (n is an integer)

To prove true for $n = 1$

$$2^1 + (-1)^{1+1} = 2 + 1$$

$$= 3 \times 1 \quad \textbf{Line 1}$$

Result is true for $n = 1$

Step 2: Assume true for $n = k$

$$\text{ie. } 2^k + (-1)^{k+1} = 3m \text{ (} m \text{ is an integer)} \quad \textbf{Line 2}$$

Step 3: To prove true for $n = k + 1$

$$2^{k+1} + (-1)^{k+1+1} = 2(2^k) + (-1)^{k+2} \quad \textbf{Line 3}$$

$$= 2[3m + (-1)^{k+1}] + (-1)^{k+2} \quad \textbf{Line 4}$$

$$= 2 \times 3m + 2 \times (-1)^{k+2} + (-1)^{k+2}$$

$$= 3[2m + (-1)^{k+2}]$$

Which is a multiple of 3 since m and k are integers.

Step 4: True by induction

In which line did Emma make an error?

- (A) Line 1
- (B) Line 2
- (C) Line 3
- (D) Line 4

Section II: 60 marks

Attempt Questions 11 – 14

Allow about 1 hour and 45 minutes for this section

Answer each question in a separate writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) *Start a new writing booklet*

- (a) Consider the function $f(x) = x^2 - 4x + 6$.
- (i) Explain why the domain of $f(x)$ must be restricted if $f(x)$ is to have an inverse function. 1
- (ii) Given that the domain of $f(x)$ is restricted to $x \leq 2$, find an expression for $f^{-1}(x)$. 2
- (iii) Given the restriction in part (ii), sketch $y = f^{-1}(x)$. 2
- (iv) The curve $y = f(x)$ with its restricted domain and the curve $y = f^{-1}(x)$ intersect at point P .
Find the coordinates of P . 1
- (b) Use the substitution $u = 1 + 2 \tan x$ to evaluate $\int_0^{\frac{\pi}{4}} \frac{1}{(1 + 2 \tan x)^2 \cos^2 x} dx$. 3

(Q11 continues on the next page)

(Q11 continued)

- (c) Solve the equation $\cos x - \sin x = 1$, where $0 \leq x \leq 2\pi$. **3**

- (d) The column (position) vector notation of 4 vectors is shown below.

$$P = \begin{pmatrix} -8 \\ -8 \end{pmatrix} \quad Q = \begin{pmatrix} 3 \\ 6 \end{pmatrix} \quad R = \begin{pmatrix} 1 \\ 5 \end{pmatrix} \quad S = \begin{pmatrix} -5 \\ 7 \end{pmatrix}$$

Find the column (position) vector notation of:

(i) $\overrightarrow{PQ}$ **1**

(ii) $\overrightarrow{RS}$ **1**

(iii) $-\overrightarrow{PQ} - \overrightarrow{RS}$ **1**

End of Question 11. Start a New Booklet.

Question 12 (15 marks) *Start a new writing booklet*

- (a) A particle is moving in a straight line such that its displacement (x metres) from a fixed point O after t seconds is given by $x = \cos 2t + \sqrt{3} \sin 2t$.
- (i) What is the maximum distance of the particle from O ? 2
- (ii) When is the particle first at the origin? 1
- (b) A heated metal ball is dropped into a liquid. As the ball cools, its temperature, T °C, t minutes after it enters the liquid, is given by:

$$T = 400e^{-0.05t} + 25, \quad t \geq 0$$

- (i) Find the temperature of the ball as it enters the liquid. 1
- (ii) Find the value of t if $T = 300$. Answer correct to 3 significant figures. 1
- (iii) Find the rate at which the temperature of the ball is decreasing at the instant when $t = 50$. Give your answer in °C per minute to 3 significant figures. 2
- (iv) Using the equation for temperature T in terms of t , given above, to explain why the temperature of the ball can never fall to 20°C. 1

(c) Find $\int_0^{\pi} \frac{4}{\sqrt{16-x^2}} dx$. 2

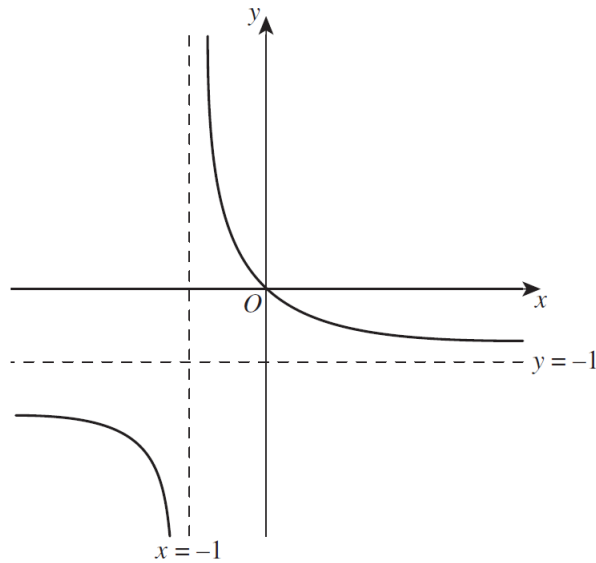
(d) (i) Use the substitution $t = \tan \frac{x}{2}$ to show that $\operatorname{cosec} x + \cot x = \cot \frac{x}{2}$. 2

(ii) Hence evaluate $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (\operatorname{cosec} x + \cot x) dx$. Answer in simplest exact form. 3

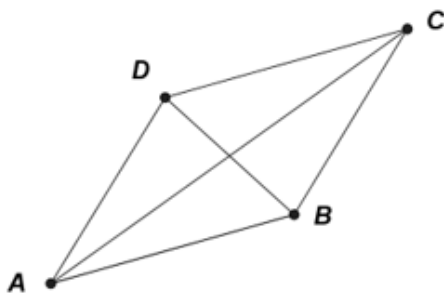
End of Question 12. *Start a New Booklet.*

Question 13 (15 marks) *Start a new writing booklet*

- (a) The diagram below is the sketch of the graph of the function $f(x) = -\frac{x}{x+1}$.



- (i) Sketch the graph of $y = (f(x))^2$, showing all asymptotes and intercepts. 2
- (ii) Solve the equation $(f(x))^2 = f(x)$. 1
- (b) $ABCD$ is a rhombus with $\overrightarrow{AB} = \mathbf{a}$ and $\overrightarrow{AD} = \mathbf{d}$.



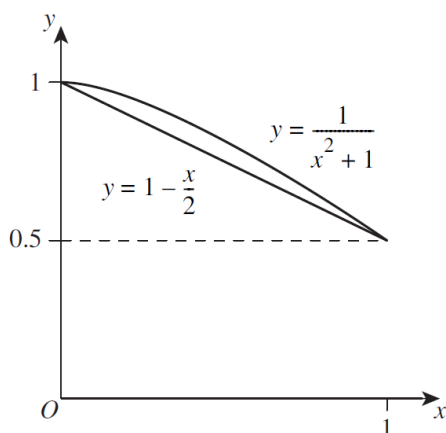
Use vector methods to prove that the diagonals of the rhombus are perpendicular to each other.

2

(Q13 continues on the next page)

(Q13 continued)

- (c) The diagram shows the graph of $y = \frac{1}{x^2 + 1}$ and the graph of $y = 1 - \frac{x}{2}$ for $0 \leq x \leq 1$.



- (i) Find the exact volume of the solid of revolution formed when the region bounded by the graph of $y = \frac{1}{x^2 + 1}$, the y -axis and the line $y = \frac{1}{2}$ is rotated about the y -axis. 2
- (ii) Find the exact volume of the solid of revolution formed when the region bounded by the graph of $y = 1 - \frac{x}{2}$, the y -axis and the line $y = \frac{1}{2}$ is rotated about the y -axis. 2
- (iii) Use the results from parts (c)(i) and (c)(ii) to show that $\frac{2}{3} < \ln 2$. 1
- (d) A multiple-choice test contains ten questions. Each question has four choices for the correct answer. Only one of the choices is correct.
- (i) What is the probability of getting 70% correct with random guessing? 1
- (ii) What is the probability of getting at most 70% correct with random guessing? 2
- (e) A binomial random variable X has a mean of 15 and a variance of 10.
What are the parameters n and p ? 2

End of Question 13. *Start a New Booklet.*

Question 14 (15 marks) *Start a new writing booklet*

- (a) Prove by mathematical induction that, for all integers $n \geq 1$,

$$\frac{2}{1 \times 3} + \frac{2}{2 \times 4} + \frac{2}{3 \times 5} + \dots + \frac{2}{n(n+2)} = \frac{3}{2} - \frac{2n+3}{(n+1)(n+2)}.$$

3

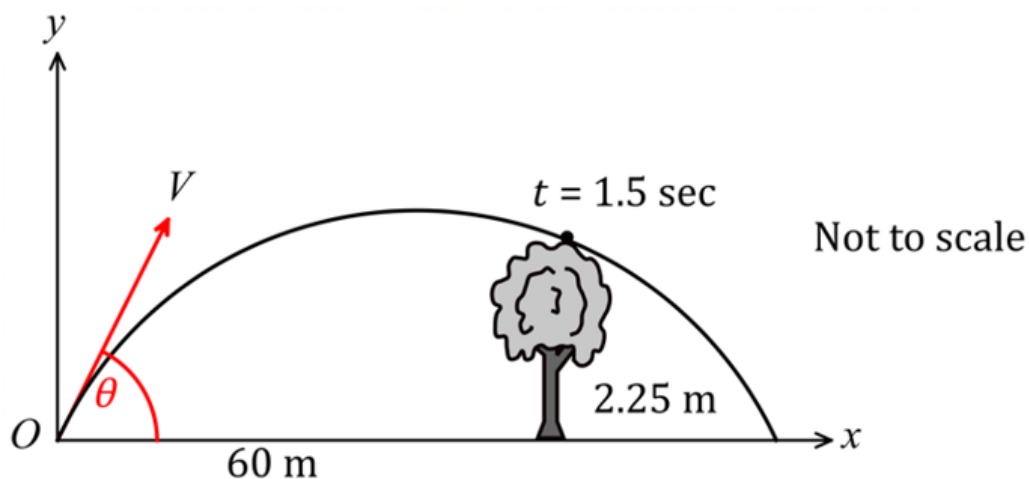
- (b) A bag contains n red marbles and one blue marble. Three marbles are drawn

(without replacement). The probability that the three marbles are red is $\frac{5}{8}$.

Find the value of n .

2

- (c) A golfer hits a golf ball from a point O with speed $V \text{ ms}^{-1}$ at an angle θ° above the horizontal, where $0 < \theta < \frac{\pi}{2}$. The ball just passes over a 2.25 m high tree after 1.5 seconds. The tree is 60 metres away from the point from which the ball was hit. Assume $g = 10 \text{ ms}^{-2}$.



- (i) What is the angle of projection of the golf ball to the nearest minute?
Assume the horizontal and vertical displacements of the golf ball are given by the vector $\underline{r}(t) = (Vt \cos \theta)\mathbf{i} + (-5t^2 + Vt \sin \theta)\mathbf{j}$.

3

- (ii) What is the initial speed ($V \text{ ms}^{-1}$) of the golf ball, correct to the nearest whole number?

1

(Q14 continues on the next page)

(Q14 continued)

- (d) The population, P , of animals in an environment in which there are scarce resources is increasing such that $\frac{dP}{dt} = P(100 - P)$, where t is time.

When $t = 0$, $P = 10$.

(i) Show that $\frac{1}{100} \left(\frac{1}{P} + \frac{1}{100 - P} \right) = \frac{1}{P(100 - P)}$. 1

(ii) Find an expression for P in terms of t . 3

- (e) The table shows selected values of a one-to-one differentiable function $g(x)$ and its derivative $g'(x)$.

x	-1	0
$g(x)$	-5	-1
$g'(x)$	3	$\frac{1}{2}$

Let $f(x)$ be a function such that $f(x) = g^{-1}(x)$.

Find the value of $f'(-1)$. 2

End of Question 14.

End of Examination.

NGS Trial Examination 2020
Year 12 Mathematics Extension 1
Worked solutions and marking guidelines

Section 1

10 marks

Question 1 (1 mark)

Solution	Answer	Mark
$(x-2)$ is a factor of $x^3 - 8x^2 + 21x - A$ $\therefore 8 - 32 + 42 - A = 0$ $A = 18$	D	1

Question 2 (1 mark)

Solution	Answer	Mark
If $y = \tan^{-1}(3x)$ $\frac{dy}{dx} = \frac{1}{1+(3x)^2} \times 3$ $= \frac{3}{1+9x^2}$	C	1

Question 3 (1 mark)

Solution	Answer	Mark
$\overrightarrow{QS} = \overrightarrow{PS} - \overrightarrow{PQ}$ $= \vec{p} - 2\vec{s}$	C	1

Question 4 (1 mark)

Solution	Answer	Mark
$T_{k+1} = {}^8C_k (x)^{8-k} \left(\frac{3}{x}\right)^k$ $= {}^8C_k x^{8-k} (3^k x^{-k})$ $= {}^8C_k (3)^k x^{8-2k}$ $x^4 \Rightarrow 8 - 2k = 4$ $k = 2$ the term is ${}^8C_2 \times (3)^2 = 252$	D	1

Question 5 (1 mark)

Solution	Answer	Mark
By inspection and properties of $y = \frac{1}{f(x)}$	B	1

Question 6 (1 mark)

Solution	Answer	Mark
By considering slopes at different points of cartesian plane and testing with each differential equation	B	1

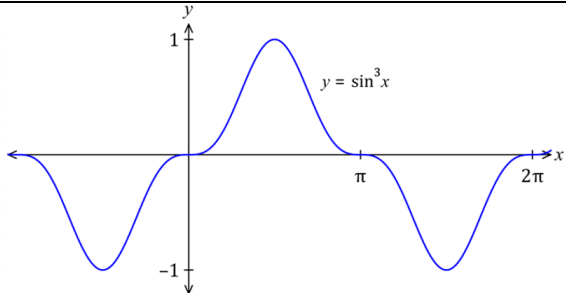
Question 7 (1 mark)

Solution	Answer	Mark
Seat females in $(4-1)!$ ways. Then seat males in $4!$ ways $\therefore$ number of ways $= 3! \times 4!$	B	1

Question 8 (1 mark)

Solution	Answer	Mark
For $y = 2 \cos^{-1} \left(\frac{x}{2} - 1 \right)$ Range $0 \leq y \leq 2\pi$ Domain is $D: -1 \leq \frac{x}{2} - 1 \leq 1$ $0 \leq x \leq 4$	A	1

Question 9 (1 mark)

Solution	Answer	Mark
 There are 3 equivalent areas from $x = -\pi$ to $x = 2\pi$ $u = \cos x$ $\frac{du}{dx} = -\sin x$ $du = -\sin x dx$ $x = 0, u = 1 \text{ and } x = \pi, u = -1$ $A = 3 \times \int_0^\pi \sin^3 x dx$ $= -3 \int_0^\pi (1 - \cos^2 x) \times -\sin x dx$ $= -3 \int_1^{-1} (1 - u^2) du$ $= 3 \int_{-1}^1 (1 - u^2) du$	D	1

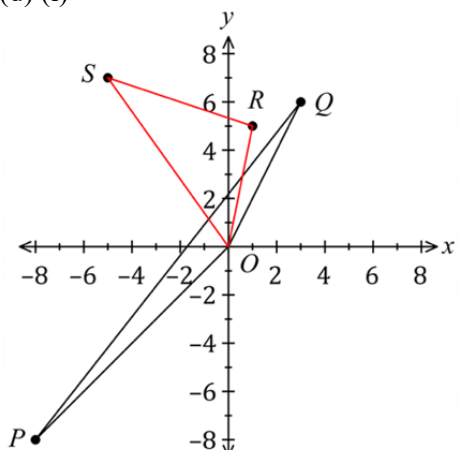
Question 10 (1 mark)

Solution	Answer	Mark
Step 3: To prove true for $n = k + 1$ $2^{k+1} + (-1)^{k+1+1} = 2(2^k) + (-1)^{k+2}$ $= 2[3m - (-1)^{k+1}] + (-1)^{k+2}$ Error Line 4 $= 2 \times 3m - 2 \times (-1)^{k+1} - (-1)^{k+1}$ $= 3[2m - (-1)^{k+1}]$ Which is a multiple of 3 since m and k are integers. Step 4: True by induction	D	1

Q11 (15 mark)

Solution	Mark (Guide only)
(a) (i) $f(x) = x^2 - 4x + 6$ is a parabola. Excluding the turning point at $(2, 2)$, for each value of $f(x)$ in the range there are two x-values. Geometrically, this corresponds to a horizontal line intersecting the graph twice. If x and y are swapped, each x-value in the domain will have two y-values. Hence the inverse will not be a function.	1 Mark: Explains using the horizontal line test or equivalent merit.
(ii) Use the completing the square method to express $f(x)$ in turning point form: $f(x) = x^2 - 4x + 6$ $= (x - 2)^2 + 2 \quad (x \geq 2)$ Swap x and y, then make y the subject. $x = (y - 2)^2 + 2$ $x - 2 = (y - 2)^2$ $y - 2 = -\sqrt{x - 2} \quad (\sqrt{x - 2} \text{ is discarded as } y \leq 2)$ $y = -\sqrt{x - 2} + 2$ $f^{-1}(x) = -\sqrt{x - 2} + 2 \quad (x \geq 2)$	2 Marks: Correct Answer 1 Mark: Swaps x and y OR equivalent merit.
(iii)	2 Marks: Correct shape and start at $(2, 2)$ 1 Mark: Correct shape OR starting point.
(iv) The curves $y = f(x)$ and $y = f^{-1}(x)$ have a common intersection with the line $y = x$. For example, attempting to solve $f(x) = x$ for x: $x^2 - 4x + 6 = x$ $x^2 - 5x + 6 = 0$ $x = 2, 3$ When $x = 2$, $y = 2$ and so $(2, 2)$ lies on the line $y = x$. When $x = 3$, $y = 1$ and so $(3, 1)$ does not lie on the line $y = x$. Therefore the coordinates of P are $(2, 2)$.	1 Mark: Correct solution

(b) Let $u = 1 + 2 \tan x$. $\frac{du}{dx} = 2 \sec^2 x = \frac{2}{\cos^2 x} \Rightarrow dx = \frac{\cos^2 x}{2} du$ When $x = 0$, $u = 1$ and when $x = \frac{\pi}{4}$, $u = 3$. $\int_0^{\frac{\pi}{4}} \frac{1}{(1 + 2 \tan x)^2 \cos^2 x} dx = \int_1^3 \frac{1}{2u^2} du$ $= -\left[\frac{1}{2u}\right]_1^3$ $= -\left(\frac{1}{6} - \frac{1}{2}\right)$ $= \frac{1}{3}$	3 Marks: Correct solution 2 Marks: Finds expression for integral in terms of u, or equivalent merit. 1 Mark: Derives $u=1+2\tan x$ correctly
(c) Substituting $\cos x = \frac{1-t^2}{1+t^2}$, $\sin x = \frac{2t}{1+t^2}$ where $t = \tan \frac{1}{2}x$ into $\cos x - \sin x = 1$ and expressing $1 = \frac{1+t^2}{1+t^2} \text{ gives:}$ $\frac{1-t^2}{1+t^2} - \frac{2t}{1+t^2} = \frac{1+t^2}{1+t^2}$ $\frac{1-t^2-2t-1-t^2}{1+t^2} = 0$ $\frac{-2(t^2+t)}{1+t^2} = 0$ $t^2+t=0$ $t(t+1)=0$ $t=-1, 0$ $\tan \frac{1}{2}x = -1, 0$ $\tan \frac{1}{2}x = 0 \Rightarrow \frac{1}{2}x = 0, \pi$ $\tan \frac{1}{2}x = -1$ $\tan$ is negative in the second quadrant and the related angle is $\frac{\pi}{4}$. $\tan \frac{1}{2}x = -1 \Rightarrow \frac{x}{2} = \frac{3\pi}{4}$ So $x = 0, \frac{3\pi}{2}, 2\pi$.	3 Marks: Correct solution 2 Marks: Determines that $\tan(\frac{1}{2}x) = -1, 0$ 1 Mark: Attempts to form a quadratic equation in t with some correct working OR equivalent merit.

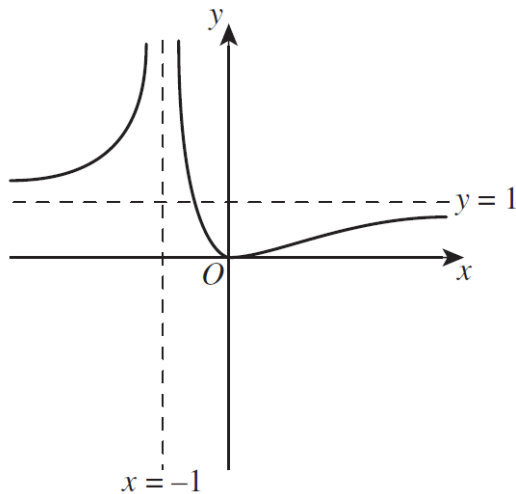
(d) (i)  $\overrightarrow{PQ} = \begin{pmatrix} 11 \\ 14 \end{pmatrix}$	1 Mark: Correct answer.
(ii) $\overrightarrow{RS} = \begin{pmatrix} -6 \\ 2 \end{pmatrix}$	1 Mark: Correct answer.
(iii) $-\overrightarrow{PQ} - \overrightarrow{RS} = \begin{pmatrix} -5 \\ -16 \end{pmatrix}$	1 Mark: Correct answer.

Q12 (15 mark)	
(a) (i) Using auxiliary angle: $x = \cos 2t + \sqrt{3} \sin 2t = 2 \cos \left(2t - \frac{\pi}{3} \right)$ $\therefore$ maximum distance from $O = 2$ metres	2 marks: Correct Answer 1 Mark: Some attempt at Auxiliary Angle with some correct working OR equivalent merit
(ii) $2 \cos \left(2t - \frac{\pi}{3} \right) = 0$ $2t - \frac{\pi}{3} = \frac{\pi}{2}$ $t = \frac{5\pi}{12} \text{ seconds}$	1 Mark: Correct answer.
(b) (i) Ball enters the liquid when $t = 0$ $T = 400e^{-0.05t} + 25$ $= 400e^{-0.05 \times 0} + 25 = 425^\circ\text{C}$	1 Mark: Correct answer.
(ii) $300 = 400e^{-0.05t} + 25$ $e^{-0.05t} = \frac{275}{400}$ $-0.05t = \ln \left(\frac{11}{16} \right)$ $t = 7.4938 \dots \approx 7.49 \text{ min (3 sig. fig.)}$	1 Mark: Correct answer.
(iii) $T = 400e^{-0.05t} + 25$ $\frac{dT}{dt} = -20e^{-0.05t}$ $= -20e^{-0.05 \times 50}$ $= -1.6416 \dots \approx -1.64^\circ\text{C/min}$ $\therefore$ Rate of decrease is 1.64°C per minute.	2 Marks: Correct answer. 1.5 marks: Only giving $-1.64/\text{min}$ 1 Mark: Differentiates correctly to find the rate of change.
(iv) When t approaches infinity then $e^{-0.05t} \rightarrow 0$ $\therefore T > 25$ and can never fall to 20°C.	1 Mark: Correct answer. (Must <u>use the equation</u> to show correctly)
(c)	

$\int_0^{\pi} \frac{4}{\sqrt{16-x^2}} dx = 4 \left[\sin^{-1} \left(\frac{x}{4} \right) \right]_0^{\pi}$ $= 4 \left[\sin^{-1} \left(\frac{\pi}{4} \right) - \sin^{-1}(0) \right]$ $= 4 \left[\frac{1}{\sqrt{2}} - 0 \right]$ $= \frac{4}{\sqrt{2}} = 2\sqrt{2}$	2 Marks: Correct answer. 1 Mark: Finds the correct integration.
(d) (i) $\text{LHS} = \operatorname{cosec} x + \cot x$ $= \frac{1+t^2}{2t} + \frac{1-t^2}{2t}$ $= \frac{1+t^2+1-t^2}{2t}$ $= \frac{1}{t}$ $= \cot \frac{x}{2}$ $= \text{RHS}$	2 Marks: Correct answer. 1 Mark: Writes cosec x and cot x in terms of t.
(ii) $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (\operatorname{cosec} x + \cot x) dx = \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cot \frac{x}{2} dx$ $= 2 \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{0.5 \cos \frac{x}{2}}{\sin \frac{x}{2}} dx$ $= 2 \left[\ln \left(\sin \frac{x}{2} \right) \right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$ $= 2 \left[\ln \left(\sin \frac{\pi}{4} \right) - \ln \left(\sin \frac{\pi}{6} \right) \right]$ $= 2 \left(\ln \frac{1}{\sqrt{2}} - \ln \frac{1}{2} \right)$ $= 2 \ln \left(2^{-\frac{1}{2}} \div 2^{-1} \right)$ $= 2 \ln 2^{\frac{1}{2}}$ $= \ln 2$	3 Marks: Correct answer. 2 Marks: Makes significant progress. <u>OR</u> uses logs correctly 1 Mark: Finds the primitive function.

Q13 (15 mark)

(a) (i)

2 Marks: Sketches correct graph with asymptotes at $x=-1$ and $y=1$ 1 Mark: Shows min turning point at origin
OR equivalent merit.

(ii)

$$(f(x))^2 = f(x) \Rightarrow f(x)(f(x) - 1) = 0$$

So $f(x) = 1$ or $f(x) = 0$.

$$-\frac{x}{x+1} = 1 \Rightarrow x = -\frac{1}{2}$$

Hence $x = -\frac{1}{2}$ or $x = 0$.**OR**The graphs of $y = f(x)$ and $y = (f(x))^2$ intersect at O , where $x = 0$.The graphs of $y = f(x)$ and $y = (f(x))^2$ intersect on the line $y = 1$, where $x = -\frac{1}{2}$.

1 Mark: Correct solution

(b)

$$\overrightarrow{AC} = \mathbf{a} + \mathbf{d}, \overrightarrow{DB} = \mathbf{a} - \mathbf{d}$$

$$\overrightarrow{AC} \cdot \overrightarrow{DB} = (\mathbf{a} + \mathbf{d}) \cdot (\mathbf{a} - \mathbf{d})$$

$$= \mathbf{a} \cdot \mathbf{a} - \mathbf{d} \cdot \mathbf{d}$$

$$= |\mathbf{a}|^2 - |\mathbf{d}|^2$$

$$= 0 \text{ (sides of rhombus have equal length)}$$

$$\text{But } \overrightarrow{AC} \cdot \overrightarrow{DB} = |\mathbf{a}| |\mathbf{d}| \cos \theta \text{ (where } \theta \text{ is the angle between the vectors)}$$

$$\therefore \cos \theta = 0$$

$$\theta = 90^\circ$$

i.e. diagonals are perpendicular

2 marks: Correct Answer.

1 mark: Writes $\overrightarrow{AC} = \mathbf{a} + \mathbf{d}$

(c)

2 marks: Correct Answer.

1 mark: Gives correct integral for volume of revolution

(i) Rearranging $y = \frac{1}{x^2 + 1}$ to express x^2 in terms of y gives $x^2 = \frac{1}{y} - 1$. $V = \pi \int_{\frac{1}{2}}^1 \left(\frac{1}{y} - 1 \right) dy$ $= \pi \left[\ln y - y \right]_{\frac{1}{2}}^1$ $= \pi \left(\ln 1 - 1 - \left(\ln \frac{1}{2} - \frac{1}{2} \right) \right)$ $= \pi \left(\ln 2 - \frac{1}{2} \right)$	
(ii) Rearranging $y = 1 - \frac{x}{2}$ to express x in terms of y gives $x = 2(1 - y)$. $V = \pi \int_{\frac{1}{2}}^1 (4(1 - y)^2) dy$ $= -\frac{4\pi}{3} \left[(1 - y)^3 \right]_{\frac{1}{2}}^1$ $= -\frac{4\pi}{3} \left(0 - \frac{1}{8} \right)$ $= \frac{\pi}{6}$ Alternatively: The solid formed is a cone of radius 1 and height $\frac{1}{2}$. Substituting these values into $V = \frac{1}{3}\pi r^2 h$ gives: $V = \frac{1}{3} \times \pi \times 1^2 \times \frac{1}{2}$ $= \frac{\pi}{6}$	2 marks: Correct Answer. 1 mark: Gives correct integral for volume of revolution
(iii) From the diagram, it can be reasoned that $\pi \left(\ln 2 - \frac{1}{2} \right) > \frac{\pi}{6}$. So $\ln 2 - \frac{1}{2} > \frac{1}{6} \Rightarrow \ln 2 > \frac{2}{3}$.	1 Mark: Correct answer.
(d) (i) Let p be the probability of getting the correct answer. $p = \frac{1}{4}, n = 10$ $P(X = x) = {}^{10}C_x \left(\frac{1}{4} \right)^x \left(\frac{3}{4} \right)^{10-x}$ $P(X = 7) = {}^{10}C_7 \left(\frac{1}{4} \right)^7 \left(\frac{3}{4} \right)^{10-7}$ $= \frac{405}{131072}$	1 Mark: Correct answer.

(ii) $P(X \leq 7) = 1 - (P(8) + P(9) + P(10))$ $= 1 - \left({}^{10}C_8 \left(\frac{1}{4}\right)^8 \left(\frac{3}{4}\right)^{10-8} + {}^{10}C_9 \left(\frac{1}{4}\right)^9 \left(\frac{3}{4}\right)^{10-9} + {}^{10}C_{10} \left(\frac{1}{4}\right)^{10} \left(\frac{3}{4}\right)^{10-10} \right)$ $= 0.99958\dots$	2 Marks: Correct answer. 1 Mark: Uses the complementary event or shows some understanding.
(e) $E(X) = np = 15 \text{ (1)}$ $Var(X) = np(1 - p) = 10 \text{ (2)}$ Substituting equation (1) into (2) $15 \times (1 - p) = 10$ $1 - p = \frac{10}{15} = \frac{2}{3} \text{ or } p = \frac{1}{3}$ Substituting $p = \frac{1}{3}$ into equation (1) $n \times \frac{1}{3} = 15$ $n = 45$ $\therefore$ Parameters are $n = 45$ and $p = \frac{1}{3}$	2 Marks: Correct answer. 1 Mark: Finds one of the parameters or shows some understanding.

Q14 (15 mark) (a) Consider $n = 1$. $\text{LHS} = \frac{2}{1 \times 3} = \frac{2}{3} \text{ and}$ $\text{RHS} = \frac{3}{2} - \frac{2(1)+3}{(1+1)(1+2)} = \frac{4}{6} = \frac{2}{3} = \text{LHS.}$ The statement is true when $n = 1$. Suppose true for $n = k$. So $\frac{2}{1 \times 3} + \frac{2}{2 \times 4} + \frac{2}{3 \times 5} + \dots + \frac{2}{k(k+2)} = \frac{3}{2} - \frac{2k+3}{(k+1)(k+2)}$. Show it is true for $n = k + 1$; that is, $\frac{2}{1 \times 3} + \frac{2}{2 \times 4} + \frac{2}{3 \times 5} + \dots + \frac{2}{k(k+2)} + \frac{2}{(k+1)(k+3)} =$ $\frac{3}{2} - \frac{2(k+1)+3}{((k+1)+1)((k+1)+2)}$ $\text{LHS} = \frac{2}{1 \times 3} + \frac{2}{2 \times 4} + \frac{2}{3 \times 5} + \dots + \frac{2}{k(k+2)} + \frac{2}{(k+1)(k+3)}$ $= \frac{3}{2} - \frac{2k+3}{(k+1)(k+2)} + \frac{2}{(k+1)(k+3)}$ $= \frac{3}{2} - \frac{(2k+3)(k+3) - 2(k+2)}{(k+1)(k+2)(k+3)}$ $= \frac{3}{2} - \frac{2k^2 + 7k + 5}{(k+1)(k+2)(k+3)}$ $= \frac{3}{2} - \frac{(2k+5)(k+1)}{(k+1)(k+2)(k+3)}$ $= \frac{3}{2} - \frac{2k+5}{(k+2)(k+3)}$ $= \frac{3}{2} - \frac{2(k+1)+3}{((k+1)+1)((k+1)+2)}$ $= \text{RHS}$ If true for $n = k$, then true for $n = k + 1$. Hence, by mathematical induction, true for $n \geq 1$.	3 Marks: Correct proof 2 Marks: Establishes the inductive step OR equivalent merit. 1 Mark: Establishes the $n=1$ case or equivalent merit.
(b) Prob(3 red marbles drawn) = $\frac{{}^nC_3}{{}^{n+1}C_3}$ ↓ $\frac{n}{n+1} \times \frac{n-1}{n} \times \frac{n-2}{n-1} = \frac{5}{8}$ $\frac{n-2}{n+1} = \frac{5}{8}$ $n = 7$ ↓ $\frac{{}^nC_3}{{}^{n+1}C_3} = \frac{5}{8}$ $\frac{\left(\frac{n(n-1)(n-2)}{3 \times 2 \times 1} \right)}{\frac{(n+1)(n)(n-1)}{3 \times 2 \times 1}} = \frac{5}{8}$ $\frac{n-2}{n+1} = \frac{5}{8}$ $8n - 16 = 5n + 5$ $n = 7$	2 marks: Correct Answer 1 mark: Writes correct answer for probability

(c) (i) After 1.5 seconds $x = 60$ and $y = 2.25$ $60 = 1.5V\cos\theta$ $V\cos\theta = 40 \text{ (1)}$ $2.25 = -5 \times 1.5^2 + 1.5V\sin\theta$ $13.5 = 1.5V\sin\theta$ $V\sin\theta = 9 \text{ (2)}$ Dividing the two equations $\frac{V\sin\theta}{V\cos\theta} = \frac{9}{40}$ $\tan\theta = \frac{9}{40}$ $\theta = \tan^{-1} \frac{9}{40} = 12^\circ 41'$ $\therefore$ Golf ball has an angle of projection of $12^\circ 41'$.	3 marks: Correct answer. 2 marks: Makes significant progress. 1 mark: Sets up the two equations or shows some understanding.
(ii) Using equations (1) and (2) $(V\sin\theta)^2 + (V\cos\theta)^2 = 9^2 + 40^2$ $V^2(\sin^2\theta + \cos^2\theta) = 9^2 + 40^2$ $V = \sqrt{9^2 + 40^2}$ $V = 41 \text{ ms}^{-1}$ $\therefore$ Speed of the gold ball is 41 ms^{-1}	1 Mark: Correct answer.
(d) (i) $\frac{1}{100} \left(\frac{1}{P} + \frac{1}{100-P} \right) = \frac{1}{100} \left(\frac{100-P+P}{P(100-P)} \right) = \frac{1}{P(100-P)}$	1 mark: Correctly shows the result

(ii) $\frac{dP}{dt} = P(100 - P)$ $\frac{1}{P(100 - P)} dP = dt$ $\int \frac{1}{P(100 - P)} dP = \int dt$ $\int \frac{1}{100} \left(\frac{1}{P} + \frac{1}{100 - P} \right) dP = \int dt$ $\int \left(\frac{1}{P} + \frac{1}{100 - P} \right) dP = 100 \int dt$ $\log_e P - \log_e (100 - P) = 100t + c$ $t = 0, P = 10 \Rightarrow \log_e 10 - \log_e 90 = c$ $c = \log_e \frac{1}{9}$ $\log_e \left(\frac{P}{100 - P} \right) = 100t + \log_e \frac{1}{9}$ $\left(\frac{P}{100 - P} \right) = e^{100t + \log_e \frac{1}{9}}$ $\frac{P}{100 - P} = \frac{1}{9} e^{100t} \quad \Rightarrow \quad \frac{100 - P}{P} = 9e^{-100t}$ $9P = 100e^{100t} - Pe^{100t}$ $9P + Pe^{100t} = 100e^{100t}$ $P = \frac{100e^{100t}}{9 + e^{100t}}$	3 marks: Correct answer 2 marks: Integrates correctly without finding the constant <u>OR</u> Integrates correctly, finds c, but leaves as $t = f(P)$. 1 mark: Makes some progress. I.e. separates the differential equation and attempts to integrate.
(e) From the table, $f(x) = g^{-1}(x)$ and so $f(-1) = g^{-1}(-1) = 0$. $f'(-1) = \frac{1}{g'(f(-1))}$ $= \frac{1}{g'(0)}$ $= \frac{1}{\frac{1}{2}}$ $= 2$	3 marks: Gives the correct solution 2 marks: Determines $f(-1) = g^{-1}(-1) = 0$ AND $f'(-1) = \frac{1}{g'(f(-1))} \dots\dots\dots$ 1 mark: Determines $f(-1) = g^{-1}(-1) = 0$ OR $f'(-1) = \frac{1}{g'(f(-1))} \dots\dots\dots$